Simpson's Paradox — Corpus Summary

A digital mirror of Simpson's Paradox — the paradox falls out of the DAG as a derived fact, never modeled directly.

Generated 2026-07-13 from live vw_* views — same data as /conclusions. SSoT is the rulebook JSON.

Scope boundary (conc-09)

This rulebook is a high-fidelity measurement instrument, not an oracle of truth. It classifies allocation geometry precisely and repeatably — sign flips, distortion ratios, signal purity — the same way a thermometer measures temperature. It does not adjudicate causal truth. Causal interpretation is external, gated by AdjustmentAppropriate, and rests on CausalRole, the one hand/LLM-encoded field in the DAG. Berkeley: highest AllocationDistortion (0.193) but CausalRole=contested — geometry and confounding are not the same thing, and the instrument only measures the former.

Scope & limits

Theorem conclusions follow from definitions (regression checks for the transpiler). Domain conclusions and corpus discovery findings are statistics on a curated convenience sample — directional, not inferential. See conc-14 before treating cross-domain patterns as laws.

Allocation sweep: 10 steps per study, f " [0.05, 0.95] (SweepStudyConfig). Latent flip = Type D at observed allocation + PooledGapCrossesZero somewhere in that range — not necessarily at observed allocation.

How to read "InvariantChecks" (read before the pass count)

This instrument is a measurement device, not an oracle: it classifies allocation geometry precisely and repeatably, the way a thermometer classifies temperature — it does not adjudicate whether a study's causal story is correct. "Invariant" in this rulebook means "the transpiler output is self-consistent with its own definitions," not "an independent empirical hypothesis survived a test." Three different things get called an invariant here, and conflating them is the single easiest way to overstate this corpus:

1. Definitional (cannot fail by construction). DistortionType, IsSignFlip, AllocationDirection, and SignalPurity are all calculated from the same three numbers (WeightedStratumGapSum, SignedPooledGap, CorrectedGap, AllocationDistortion). An invariant like "DistortionType " {A,B} !' IsSignFlip=TRUE" restates one algebraic fact in a second vocabulary — it is a regression check that the transpiler did not corrupt a formula, not a finding about the world. Most of the 32 rows in InvariantChecks are this kind.

2. Judgment-crossing (the closest thing to a substantive check). CausalRole on StratumVariables (confounder / mediator / collider / selection / contested / unknown) is the one hand/LLM-encoded field in the DAG — schema type "raw," not derived from geometry. Invariants that reference StratumCausalRole, ConfounderIdentity, or IsParadoxExplained (e.g. inv-explained-sign-flip-confounder, inv-collider-no-manifest) are checking that the geometric result is consistent with that human judgment call — the only place a real counterexample could show up. The rulebook does not currently record who made each CausalRole call or by what protocol; treat these classifications as reviewed labels, not as independently verified ground truth.

3. Vacuous (PassCount=0). A handful of invariants (e.g. inv-type-a-unanimous, inv-confounder-signflip-explained, inv-collider-no-manifest) currently have zero rows in their filtered universe. "0 pass / 0 fail" means no study in this corpus falls into that category yet — it is true the way "all unicorns are blue" is true, and proves nothing until a qualifying study exists.

Epistemic summary (tiered)

Proved by construction (pure algebra — no CausalRole dependency): 4
Proved, conditional on CausalRole annotation (see caveats below — not pure algebra): 4
Established in instrument (instrument · taxonomy · methodology · scope): 24
Observed in this corpus (domain — provisional): 2
Total conclusions: 34
Corpus hypotheses (loop-61): 23 / 25 confirmed
Consistency checks (definition-linked): 9 / 9 pass
Latent Type-D (under sweep contract above): 112 / 138 (81.2%)

Studies in corpus: 242
Distortion taxonomy: A 79 · B 10 · C+ undefined · C" undefined · D 138

Pre-registered discovery (loop-61)

Consistency checks — confirm implementation matches stated algebra (same tier as conc-12 / inv-signal-purity-sign-flip). Not independent discoveries about nature.

H-collider-no-manifest-theorem — PASS (consistency check)

At scale ("e10 collider/selection studies post expansion), zero manifest sign-flips: ColliderSelectionManifestCount = 0.

Expected: ColliderSelectionManifestCount = 0 AND ColliderSelectionCount >= 10
Observed: collN=11 collManifest=0

H-corrected-gap-invariant — PASS (consistency check)

CorrectedGap is allocation-invariant for every study: MAX(SweepStudySummary.SweepCorrectedGapRange) < 0.0001 and zero failing studies.

Expected: MaxStudySweepCorrectedGapRange < 0.0001 AND CorrectedGapInvariantFailCount = 0
Observed: maxRange=0.000 fails=0

H-explained-bidirectional — PASS (consistency check)

IsParadoxExplained 'ú confounder manifest sign-flip: ExplainedConfounderCount = ConfounderSignFlipCount, no false-positive explained rows, no unexplained confounder sign-flips.

Expected: ExplainedConfounderCount = ConfounderSignFlipCount AND FalsePositiveExplainedCount = 0 AND UnexplainedConfounderSignFlipCount = 0
Observed: explained=85 confFlip=85 falsePos=0 unexplained=0

H-identity-map-coverage — PASS (consistency check)

At least 95% of real-study StratumVariables carry a ConfounderIdentity map — ontology covers the corpus.

Expected: IdentityMapCoverageRate >= 0.95
Observed: IdentityMapCoverageRate=1.000

H-perfect-balance-sufficient-safety — PASS (consistency check)

Perfect allocation balance (ArmSizeRatio=0.5 AND MaxStratumImbalance=0.0) is a domain-agnostic sufficient condition for no manifest Simpson's Paradox — no such study produces IsSignFlip=TRUE regardless of mechanism class, domain, or confounder identity.

Expected: PerfectBalanceFlipCount=0 AND PerfectBalanceStudyCount>=100
Observed: PerfectBalanceFlipCount=0 PerfectBalanceStudyCount=118

H-purity — PASS (consistency check)

Every sign-flip study has SignalPurity strictly below 0.5.

Expected: SignFlipSignalPurityMax < 0.5
Observed: maxPurity=0.500

H-purity-inversion — PASS (consistency check)

Mechanism classes with PolicyDefault='do-not-condition' or 'do-not-naively-adjust' (collider-selection, selection-frailty) have strictly higher mean SignalPurity than all stratify-* mechanism classes. High SignalPurity anti-predicts safe stratification — it is a trap for researchers who use data cleanliness as a quality heuristic.

Expected: MeanPurityColliderSelection > MeanPurityAllStratifyClasses AND MeanPuritySelectionFrailty >= 0.95
Observed: MeanPurityDoNotCondition=0.908 MeanPurityStratifyClasses=0.728

H-theorem-portfolio — PASS (consistency check)

Four Category=theorem conclusions are witnessed: SignalPurity (conc-12/20) plus three new theorems from loops 73–75.

Expected: TheoremCount >= 4
Observed: theoremCount=8

H-unexplained-flips-only-nonconfounders — PASS (consistency check)

All unexplained sign-flips (IsSignFlip=TRUE AND IsParadoxExplained=FALSE) have ConfounderIdentity " {id-contested-org-choice, id-collider-proxy} OR CausalRole=contested — no unexplained flip belongs to an unambiguous confounder identity.

Expected: UnexplainedConfounderSignFlipCount = 0
Observed: UnexplainedConfounderSignFlipCount=0

Corpus hypotheses — contingent patterns on this convenience sample. Directional inequalities only; not inferential. See conc-14.

H-age-identity-flip-rate — PASS (corpus hypothesis)

Age-composition identity (id-age-composition) produces manifest flip rate "e 40% among real studies — cross-domain demographic confounding is flip-prone, not latent-only.

Expected: AgeIdentityManifestFlipRate >= 0.40 AND AgeIdentityStudyCount >= 10
Observed: AgeIdentityManifestFlipRate=0.486 AgeIdentityStudyCount=35

H-catalog-exact-match — PASS (corpus hypothesis)

Catalog ExpectedDistortionType exact-match rate is below 50% — curator tags behave as mechanism priors, not taxonomy predictors.

Expected: TypePredictionMatchRate < 0.5
Observed: exactRate=0.391 (92/235)

H-causal-latent — PASS (corpus hypothesis)

Collider/selection studies produce latent-only flips under allocation sweep without manifest sign-flips at observed allocation.

Expected: ColliderSelectionManifestCount = 0 AND ColliderSelectionLatentOnlyCount >= 5
Observed: collManifest=0 collLatent=10 collN=11

H-causal-manifest — PASS (corpus hypothesis)

Confounder-labeled studies produce manifest sign-flips at observed allocation (IsSignFlip=TRUE) far more often than collider/selection studies — reversal mode differs by mechanism class.

Expected: ConfounderSignFlipCount > ColliderSelectionManifestCount AND ConfounderSignFlipCount >= 10
Observed: confFlip=85 collManifest=0

H-collider-identity-low-manifest — PASS (corpus hypothesis)

Collider-proxy identity (id-collider-proxy) produces "d15% manifest flip rate — near-zero manifest, predominantly latent Type-D geometry.

Expected: ManifestFlipRate(cluster-id-collider-proxy) <= 0.15 AND StudyCount(cluster-id-collider-proxy) >= 5
Observed: ColliderIdentityManifestFlipRate=0.091 ColliderIdentityStudyCount=11

H-collider-no-manifest-v2 — PASS (corpus hypothesis)

Collider and selection studies produce zero manifest sign-flips (DistortionType A or B) at observed allocation — including hernandez-diaz birth-weight collider tagged Expected A in catalog.

Expected: ColliderSelectionManifestCount = 0
Observed: collManifest=0

H-confounder-drift-type-d — FAIL (corpus hypothesis)

When studies are bucketed by real-world valid-time (cohort midpoint, not publication year), case-mix / geographic / situational confounder archetypes (MechanismClass in severity-case-mix, geography, sports-situation) show DriftDirection=rising toward clean Type-D outcomes in ConfounderDistortionTimeline, while age / institutional / SES archetypes (demographic-composition, organizational-unit, ses-income) show DriftDirection in (flat, single-decade) -- i.e. the drift is confounder-specific, not corpus-wide.

Expected: COUNTIFS(ConfounderDistortionTimeline[MechanismClass],"severity-case-mix|geography|sports-situation",ConfounderDistortionTimeline[DriftDirection],"rising") = COUNTIFS(ConfounderDistortionTimeline[MechanismClass],"severity-case-mix|geography|sports-situation") AND COUNTIFS(ConfounderDistortionTimeline[MechanismClass],"demographic-composition|organizational-unit|ses-income",ConfounderDistortionTimeline[DriftDirection],"rising") = 0
Observed: severity=falling geography=flat situational=rising age=falling institutional=rising socioeconomic=rising — expected severity/geography/situational in (rising,single-decade) and age/institutional/socioeconomic <> rising; institutional and socioeconomic both drifted rising, contradicting the hypothesis.

H-control-no-manifest-flip — PASS (corpus hypothesis)

Zero manifest sign flips among control studies. These studies were selected for absence of paradox, so no pooled-vs-stratified reversal should be present.

Expected: COUNTIF(TreatmentRankings[IsSignFlip],TRUE,Studies[IsControlStudy],TRUE) = 0

Observed: ControlManifestFlipCount=0 — vacuous: control studies have zero TreatmentRankings rows (never encoded with paired arms), so this is true by construction, not a substantive corpus finding.

H-control-safety-corridor — PASS (corpus hypothesis)

No control study with MaxStratumImbalance < 0.033 (the safe-corridor threshold) has IsSignFlip=TRUE. The safety corridor hypothesis predicts zero flips below this boundary.

Expected: COUNTIF(TreatmentRankings[IsSignFlip],TRUE,TreatmentRankings[MaxStratumImbalance],"<0.033", Studies[IsControlStudy],TRUE) = 0

Observed: ControlSafeCorridorFlipCount=0 — vacuous: control studies have zero TreatmentRankings rows (never encoded with paired arms), so this is true by construction, not a substantive corpus finding.

H-control-type-d-predominance — PASS (corpus hypothesis)

At least 70% of control studies encode as Type-D (neutral allocation, no sign flip) — Type-D is the expected basin for studies selected for absence of paradox.

Expected: COUNTIF(TreatmentRankings[IsSignFlip],TRUE,Studies[IsControlStudy],TRUE) / COUNTIF(Studies[IsControlStudy],TRUE) < 0.3

Observed: ControlFlipRate=0.000 — vacuous: control studies have zero TreatmentRankings rows (never encoded with paired arms), so this is true by construction, not a substantive corpus finding.

H-cplus-magnitude — PASS (corpus hypothesis)

C+ studies carry higher mean AllocationDistortion than C- studies and Type-D studies — magnitude inflation is the largest quantitative error class and invisible to is_sign_flip.

Expected: CPlusAvgDistortion > CMinusAvgDistortion AND CPlusAvgDistortion > TypeDAvgDistortion

Observed: C+=0.111 C-=0.030 D=0.000

H-explained-confounder — PASS (corpus hypothesis)

IsParadoxExplained is TRUE for all and only confounder sign-flip studies — adjust-resolvable confounding is the unique condition that makes a reversal fully explained.

Expected: ExplainedConfounderCount = SignFlipCount AND ExplainedNonConfounderCount = 0

Observed: ExplainedConfounderCount=85 ConfounderSignFlipCount=85

H-geographic-stable-type-d — PASS (corpus hypothesis)

Geographic-composition identity (id-geographic-composition) has "e25% Type-D fraction — stable latent-D presence distinct from manifest-flip-prone identities.

Expected: TypeDFraction(cluster-id-geographic-composition) >= 0.25 AND StudyCount(cluster-id-geographic-composition) >= 3

Observed: GeographicTypeDFraction=0.647 GeographicStudyCount=17

H-latent-d — PASS (corpus hypothesis)

More than half of Type-D (SAFE) studies would flip their pooled winner under counterfactual allocation reweighting.

Expected: LatentTypeDFraction > 0.5

Observed: fraction=0.812

H-mechanism-other-drain — PASS (corpus hypothesis)

After ontology drain, id-mechanism-other contains "d5 studies — the catch-all is no longer the dominant bucket.

Expected: StudyCount(cluster-id-mechanism-other) <= 5

Observed: MechanismOtherStudyCount=0

H-purity-trap-flag-coverage — PASS (corpus hypothesis)

PurityTrapFlag=TRUE covers "e80% of collider/selection studies and "d5% of demographic-composition confounder studies — the flag correctly discriminates the trap from the safe-stratification zone.

Expected: PurityTrapCoverage(collider-selection) >= 0.80 AND PurityTrapFalsePositiveRate(demographic-composition) <= 0.05

Observed: ColliderSelectionCoverage=0.818 DemographicFalsePositiveRate=0.000

H-safety-corridor-lower-bound — PASS (corpus hypothesis)

The minimum observed MaxStratumImbalance among all manifest sign-flip studies is 0.033 (flu-vaccine-elderly-mortality). No sign-flip occurs below this threshold — the gap [0.0, 0.033) is a corpus-level safety corridor.

Expected: MinSignFlipStratumImbalance >= 0.033 AND SafeCorridorFlipCount=0

Observed: MinSignFlipStratumImbalance=0.033 SafeCorridorFlipCount=0

H-selection-frailty-zero-manifest — PASS (corpus hypothesis)

Selection-frailty identity (id-selection-frailty) produces zero manifest sign-flips — pure latent-D geometry, no allocation-reversal at observed weights.

Expected: ManifestFlipCount(cluster-id-selection-frailty) = 0 AND StudyCount(cluster-id-selection-frailty) >= 3

Observed: SelectionFrailtyManifestFlipCount=0 SelectionFrailtyStudyCount=5

H-severity-epi-latent-only — PASS (corpus hypothesis)

Disease-severity identity within epidemiology domain shows zero manifest flips — latent-only geometry in that domain.

Expected: ManifestFlipCount(cell-id-disease-severity-x-epidemiology) = 0

Observed: SeverityEpiManifestFlipCount=0

H-severity-medicine-manifest — PASS (corpus hypothesis)

Disease-severity identity within medicine domain has manifest flip rate "e45% — severity drives reversals in clinical settings.

Expected: ManifestFlipRate(cell-id-disease-severity-x-medicine) >= 0.45

Observed: SeverityMedicineManifestFlipRate=0.533

H-severity-vs-age-latent — FAIL (corpus hypothesis)

Disease-severity identity shows lower Type-D latent fraction than age-composition identity — severity drives manifest reversals more than hidden sweep sensitivity.

Expected: SeverityIdentityLatentFractionAmongTypeD < AgeIdentityLatentFractionAmongTypeD

Observed: SeverityIdentityLatentFractionAmongTypeD=0.000 AgeIdentityLatentFractionAmongTypeD=0.000

H-small-effect — PASS (corpus hypothesis)

Allocation-stable Type-D studies carry larger observed pooled gaps than latent Type-D studies.

Expected: AvgPooledGapStableD > AvgPooledGapLatentD

Observed: stable=0.113 latent=0.020

H-synthetic-edge-coverage — PASS (corpus hypothesis)

All four corpus theorem boundary regions (near-zero-distortion, equal-weight sign-flip, zero-corrected-gap, collider-purity) are covered by at least one synthetic study.

Expected: COUNTIF(Studies[IsSynthetic],TRUE) >= 4 AND all four boundary regions represented

Observed: SyntheticStudyCount=9 near-zero-distortion=t equal-weight-sign-flip=t zero-corrected-gap=t collider-purity=t

H-ultra-fragile — PASS (corpus hypothesis)

At least four Type-D studies are sweep-fragile: signal_purity=1, allocation_distortion=0, sweep_pooled_gap_range>0.3, allocation_fragility>=10 — perfectly safe by pooled metrics yet latent-flip under reweighting.

Expected: SweepFragileCount >= 4

Observed: SweepFragileCount=8

H-unexplained-nonconfounder — PASS (corpus hypothesis)

Studies with IsSignFlip=TRUE and CausalRole != confounder (contested or mediator) have IsParadoxExplained=FALSE.

Expected: ContestedOrMediatorExplainedCount = 0

Observed: ContestedOrMediatorExplainedCount=0

Proved (by construction) (8)

conc-12-signal-purity-theorem · Proved (by construction)

SignalPurity < 0.5 is a necessary condition for allocation-driven reversal

Follows directly from how SignalPurity/CorrectedGap/AllocationDistortion are defined — corpus-scale testing here is a regression check that the transpiler did not corrupt the formula, not independent empirical confirmation of a separate discovery.

Witnessed across all reversal studies (type A + B) in the 90+ study corpus: every study with AllocationDirection='reversal' has SignalPurity < 0.5. The algebraic proof: reversal requires SignedPooledGap and CorrectedGap to have opposite signs, which requires AllocationDistortion > |CorrectedGap|, which is exactly SignalPurity < 0.5. inv-signal-purity-sign-flip enforces this as a DAG invariant. AvgSignalPurityReversal "H 0.35 vs AvgSignalPurityNonReversal "H 0.65 — gap of ~0.30.

Loop replay: loop-44
Protecting invariants: 1

conc-20-signal-purity-ceiling · Proved (by construction)

SignalPurity < 0.5 is a build-breaking invariant for sign-flip studies — witnessed at corpus scale

Follows directly from how SignalPurity/CorrectedGap/AllocationDistortion are defined — corpus-scale testing here is a regression check that the transpiler did not corrupt the formula, not independent empirical confirmation of a separate discovery.

At 96-study corpus scale, every manifest sign-flip has SignalPurity strictly below 0.5 (corpus max: 0.483, jeter-justice-1997). The melanoma-altman-1991 boundary case (purity=0.479, C+, pooled gap crosses zero, no flip) confirms the law is tight but not violated. This graduates from corpus hypothesis (loop-61) to build-breaking invariant: any future study encoding that produces a sign-flip with purity >= 0.5 is an encoding error, not a counterexample.

Loop replay: loop-64

conc-28-corrected-gap-invariance-theorem · Proved (by construction)

CorrectedGap (= WeightedStratumGapSum) is invariant under allocation reweighting — the conserved quantity

Follows directly from how SignalPurity/CorrectedGap/AllocationDistortion are defined — corpus-scale testing here is a regression check that the transpiler did not corrupt the formula, not independent empirical confirmation of a separate discovery.

Loop-73: inv-corrected-gap-invariant PASS 238/238; MaxStudySweepCorrectedGapRange=0; CorrectedGapInvariantFailCount=0. CorrectedGap (=WeightedStratumGapSum) invariant under allocation reweighting across full corpus including expansion wave 3.

Loop replay: loop-73
Protecting invariants: 1

conc-29-explained-confounder-theorem · Proved, conditional on CausalRole annotation (not pure algebra)

IsParadoxExplained 'ú confounder manifest sign-flip — adjust-resolvable confounding is the unique explanation condition

Conditional on CausalRole being correctly annotated — the one hand/LLM-encoded field in the DAG, with no recorded protocol for who assigned each label or how. True given a correct annotation; not true by algebra alone. See "How to read InvariantChecks."

Loop-74: H-explained-bidirectional PASS. ExplainedConfounderCount=ConfounderSignFlipCount; FalsePositiveExplainedCount=0; UnexplainedConfounderSignFlipCount=0. inv-explained-sign-flip-confounder + inv-confounder-signflip-explained both PASS.

Loop replay: loop-74
Protecting invariants: 2

conc-30-collider-no-manifest-theorem · Proved, conditional on CausalRole annotation (not pure algebra)

Given correct causal-role annotation, collider/selection studies produce no manifest sign-flips at observed allocation

Conditional on CausalRole being correctly annotated — the one hand/LLM-encoded field in the DAG, with no recorded protocol for who assigned each label or how. True given a correct annotation; not true by algebra alone. See "How to read InvariantChecks."

Loop-75: ColliderSelectionCount>=10 with ColliderSelectionManifestCount=0 after encoding heckman-1979 + hoxby-murarka-2009 and re-tagging simonsen/hammond to selection; fryer-2016 reclassified contested. inv-collider-no-manifest PASS.

Loop replay: loop-75
Protecting invariants: 1

conc-35-unexplained-identity-sink-theorem · Proved, conditional on CausalRole annotation (not pure algebra)

Identity-conditioned explained theorem: all unexplained sign-flips confined to contested-org-choice and collider-proxy identities

Conditional on CausalRole being correctly annotated — the one hand/LLM-encoded field in the DAG, with no recorded protocol for who assigned each label or how. True given a correct annotation; not true by algebra alone. See "How to read InvariantChecks."

Loop-85 at N=238: H-unexplained-flips-only-nonconfounders PASS. UnexplainedConfounderSignFlipCount=0. The three unexplained flips (conc-29 exception list) map exclusively to id-contested-org-choice (2) and id-collider-proxy (1). This extends the loop-74 biconditional (conc-29): explained!"confounder holds for every unambiguous confounder identity; violations are confined to ontologically ambiguous categories. inv-unexplained-identity-sink: PASS (critical, 30 total invariants).

Loop replay: loop-85
Protecting invariants: 1

conc-36-perfect-balance-theorem · Proved (by construction)

Perfect allocation balance is a sufficient condition for no manifest Simpson's Paradox

Loop-86: inv-perfect-balance-no-flip PASS 118/118. All 118 TreatmentRankings with ArmSizeRatio=0.5 AND MaxStratumImbalance=0.0 have IsSignFlip=FALSE across all 8 domains and 19 confounder archetypes. Minimum imbalance at any sign-flip study: 0.033 (flu-vaccine-elderly-mortality, demographic-composition). Safety corridor [0.0, 0.033) contains zero sign-flips. This is a sufficient-condition theorem: perfect balance guarantees no manifest paradox, domain-agnostically.

Loop replay: loop-86
Protecting invariants: 1

conc-37-purity-inversion-theorem · Proved, conditional on CausalRole annotation (not pure algebra)

Purity inversion: high SignalPurity anti-predicts safe stratification — collider geometry maximizes it

Conditional on CausalRole being correctly annotated — the one hand/LLM-encoded field in the DAG, with no recorded protocol for who assigned each label or how. True given a correct annotation; not true by algebra alone. See "How to read InvariantChecks."

Loop-87: collider/selection mean SignalPurity=0.916 (n=12); selection/frailty mean SignalPurity=1.000 (n=5) — the two mechanism classes that must NOT be stratified have the highest purity in the corpus. Demographic-composition (stratify-appropriate) mean SignalPurity=0.624 (n=35). inv-collider-purity-dominance PASS. H-purity-inversion confirmed. PurityTrapFlag encodes the trap for direct query. Researchers using data cleanliness as a quality heuristic will systematically misidentify collider/selection studies as…

Loop replay: loop-87
Protecting invariants: 1

Established in instrument (14)

conc-03-binary-excludes-partial · Established in instrument

The unanimity criterion (IsReversal) incorrectly excludes Berkeley 1973 despite its being a Simpson-type paradox

Berkeley: WSGSUM=" 0.0642, SignedPooledGap=+0.1292, IsSignFlip=TRUE, AllocationDistortion=0.1934 — greater distortion than kidney-1986. But IsReversal=FALSE because dept-C marginally favors male. The unanimity criterion is a strict subtype, not the primary definition.

Loop replay: loop-17
Protecting invariants: 5

conc-04-type-c-exists · Established in instrument

Allocation can compress the pooled signal without flipping it (type-C): a class invisible to IsReversal and IsSignFlip but visible to AllocationDistortion

compressed-synthetic: A wins every stratum and wins pooled, but the pooled margin is 6pp rather than the equal-weight 10pp. AllocationDistortion=0.0400, IsSignFlip=FALSE. The pooled winner is correct; the effect size is wrong by 4pp.

Loop replay: loop-19
Protecting invariants: 4

conc-05-real-data-generalizes · Established in instrument

Two real published studies (reintjes-2000, radelet-1981) hydrate as type-A full reversals without schema changes

Both studies entered via 05b-customize-data.sql. Witnessed: reintjes AllocationDistortion=0.0516, DistortionType=A; radelet AllocationDistortion=0.0684, DistortionType=A. No DAG or schema change was needed — the hydration contract generalizes.

Loop replay: loop-20b

conc-06-policy-derivable · Established in instrument

PolicyImplication is domain-neutral: the same formula produces researcher actions across medicine, law, and education

All five current studies (kidney-1986, berkeley-1973, compressed-synthetic, kidney-balanced, balanced-synthetic) plus two real studies produce unambiguous PolicyImplication values from DistortionType alone. No domain-specific knowledge is encoded in the formula.

Loop replay: loop-21

conc-07-model-self-portrait · Established in instrument

ModelSummary witnesses the instrument's own epistemic coverage as a first-class derived fact

ModelSummary aggregates ReversalCount, ExplainedCount, TypeA/B/C/D counts, and AvgAllocationDistortion across all studies. The model describes its own completeness in a single row.

Loop replay: loop-10

conc-11-reversal-recovery · Established in instrument

The model can produce the allocation-corrected winner directly from existing DAG fields — no new data required

CorrectedGap = WeightedStratumGapSum (already derived in loop-12/17). CorrectedWinner is a sign-read on CorrectedGap. CorrectedVsPooledAgreement is TRUE for Type-D studies and FALSE for Type-A/B studies. For kidney-1986: CorrectedGap=+0.0537, CorrectedWinner=A, CorrectedVsPooledAgreement=FALSE (pooled said B; corrected says A). The reversal recovery is free — it costs zero new data.

Loop replay: loop-27
Protecting invariants: 2

conc-15-latent-flip-potential · Established in instrument

Type-D at observed allocation does not imply allocation-stable — 70% of SAFE studies flip under counterfactual reweighting

Sensitivity over sweep range f " [0.05, 0.95]; not a claim about realistic reallocation alone.

LatentTypeDCount=45, TypeDCount=64, LatentTypeDFraction=0.703. LatentFlipPotential=TRUE when DistortionType=D AND PooledGapCrossesZero=TRUE. Examples: uc-systemwide-admissions (Type D, sweep range 0.56), recsys-offline-eval (Type D, sweep range 0.56). The ScreeningTier SAFE label describes observed allocation only; sweep reveals latent flip potential.

Loop replay: loop-61
Protecting invariants: 1

conc-16-small-effect-fragile · Established in instrument

In this corpus, allocation-stable Type-D rows carry larger observed pooled gaps than latent Type-D rows

AvgPooledGapStableD=0.102 across 19 stable Type-D studies; AvgPooledGapLatentD=0.022 across 45 latent Type-D studies. Stable studies also have lower sweep range (avg 0.072 vs 0.211). Corpus-descriptive pattern on loop-61 convenience sample — not a deductive law. Small observed effects tend to be allocation-fragile even when currently classified SAFE.

Loop replay: loop-61

conc-18-causal-role-flip-mode · Established in instrument

Causal role predicts manifest vs latent reversal mode — confounders flip at observed allocation; collider/selection flips stay sweep-latent

Loop-62 research sweep: H-causal-manifest PASS (confFlip=15 vs collManifest=0). H-causal-latent PASS (collManifest=0, collLatent=5, collN=7). Confounder Type-D rows often show latent-only sweep sensitivity (45 latent-only) without manifest sign-flip — the mechanism split is about reversal mode when sign-flips occur, not absence of sweep fragility. Existing LatentFlipPotential + StratumCausalRole suffice; defer ManifestReversal first-class fields to loop-67+.

Loop replay: loop-62

conc-19-explained-tracks-confounding · Established in instrument

IsParadoxExplained tracks adjust-resolvable confounding, not geometric distortion

In this corpus, IsParadoxExplained=TRUE iff the study has a manifest sign-flip driven by a confounder — i.e. stratification resolves the reversal. It is FALSE for all C+/C" /D types, and FALSE for the three contested/mediator sign-flips (berkeley-1973, berkeley-six-dept-1973, birth-weight-paradox). The boundary separates 'stratify immediately' from 'do not naively adjust for a collider or mediator'.

Loop replay: loop-63

conc-24-collider-no-manifest-v2 · Established in instrument

Collider/selection studies produce zero manifest sign-flips at observed allocation — theorem candidacy

Loop-69: H-collider-no-manifest-v2 PASS. ColliderSelectionManifestCount=0 including hernandez-diaz birth-weight collider (Expected-A catalog tag, observed Type D). Catalog Expected-A does not predict collider geometry.

Loop replay: loop-69

conc-25-cplus-magnitude-blindspot · Established in instrument

Type-C+ carries the largest mean AllocationDistortion — invisible to is_sign_flip screens

Loop-69: H-cplus-magnitude PASS. CPlusAvgDistortion > CMinusAvgDistortion and > TypeDAvgDistortion. Magnitude inflation is the largest quantitative error class among non-flip distortions.

Loop replay: loop-69

conc-26-sweep-fragile-class · Established in instrument

Sweep-fragile Type-D class: pooled-safe studies with wide counterfactual wander ("e4 in corpus)

Loop-69: H-ultra-fragile PASS. SweepFragileCount>=4 — Type-D rows with signal_purity=1, allocation_distortion=0, sweep_pooled_gap_range>0.3, allocation_fragility>=10. Invisible to pooled-only screens.

Loop replay: loop-69

conc-33-confounder-identity-layer · Established in instrument

ConfounderIdentity ontology: cross-study accountable stratum archetypes (loop-80)

Loop-81 witness at N=238: 15 identities, 233/233 real stratum variables mapped (100% coverage). H-age-identity-flip-rate PASS (48.6% manifest, n=35). H-severity-vs-age-latent FAIL globally (0.917 > 0.813) — domain modulation deferred to loop-82 IdentityDomainCells.

Loop replay: loop-81

Taxonomy (1)

conc-13-cplus-cminus-opposite-errors · Taxonomy

C+ (amplification) and C- (compression) are epistemically opposite errors with opposite clinical implications

titanic-1912 (C+, DistortionRatio=1.359): allocation inflates first-class survival advantage from 27.5pp corrected to 37.4pp pooled — overconfidence. melanoma-altman-1991 (C+, DistortionRatio=2.087): allocation doubles apparent female survival advantage — severe overconfidence. hannan-1994 (C-, DistortionRatio=0.314): allocation compresses CABG advantage from 8.9pp corrected to 2.8pp pooled — underconfidence, risk of premature treatment abandonment. The prior type-C category hid this distinction. Now separated b…

Loop replay: [loop-43]

Methodology (7)

conc-10-instrument-specifiable · Methodology

The full instrument can be specified precisely enough for any researcher to apply it to a new study

Witnessed in loop-26: InstrumentSpec table enumerates 5 input fields, 10 derived coordinates, 2 classification rules, and 3 adapter-requirement rows. Machine-verifiable against the full 90+ study corpus.

Loop replay: [loop-26]

conc-14-scale-is-discovery-path · Methodology

Cross-domain SignalPurity correlations require a pre-registered unbiased corpus of 50+ studies to distinguish pattern from selection artifact

Partial domain patterns witnessed across 90+ studies (loop-61): epidemiology avg distortion 0.050 vs education 0.004; legal/sports/epidemiology flip rates ~27% vs economics 0%. Full causal domain!'DistortionType mapping still requires pre-registered expansion beyond convenience sample, but the instrument now supports DiscoveryHypotheses/Findings for ongoing corpus-scale tests.

Loop replay: [loop-61]

conc-21-catalog-tags-not-predictors · Methodology

Catalog ExpectedDistortionType tags are mechanism priors, not taxonomy predictors — exact match rate 19.4%

Loop-65 audit: 18/93 exact matches (19.4%), 75 mismatches. Sign-flip prediction 18/71 (25.4%). Biggest bucket Expected A!'observed D (23 studies). Both H-catalog hypotheses PASS (<50%). Defer ExpectedSignFlip vs ExpectedDistortionType split to loop-67+.

Loop replay: [loop-65]

conc-23-research-sweep-synthesis · Methodology

Research sweep loops 62–66: confirmed laws vs corpus patterns vs deferred vocabulary

Laws (build-breaking): SignalPurity<0.5 for sign-flips (loop-64), IsParadoxExplained!"confounder (loop-63). Corpus patterns: causal-role reversal mode (loop-62), latent Type-D 70% (loop-61), catalog tag falsification 19% exact (loop-65), domain gap survives geometry control (loop-66). Deferred loop-67+: ManifestReversal/LatentReversal fields, ExpectedSignFlip catalog split, Discovery UI polish. inv-discovery-all-confirmed: 11/11 PASS.

Loop replay: [loop-66]
Protecting invariants: 1

conc-31-theorem-portfolio-synthesis · Methodology

Four witnessed theorems: SignalPurity, CorrectedGap invariance, Explained!"Confounder, Collider-no-manifest — plus corpus patterns that remain non-theorems

Loop-76: TheoremCount>=4 (conc-12/20 SignalPurity + conc-28/29/30). All loops 61-75 discovery hypotheses PASS. Methods-export ready. **Loop-78 update:** six corpus patterns superseded at N=238; theorems and build-breaking invariants unchanged — see conc-32.

Loop replay: [loop-76]
Protecting invariants: 1

conc-32-expansion-wave-3-supersession · Methodology

Expansion wave 3 (loop-77!'78): six corpus patterns superseded at N=238 — theorems and invariants hold

Loop-78 LoopReview at 238 studies (233 real). Six EpistemicTier=corpus-pattern-superseded hypotheses witnessed FAIL: H-econ-zero (flips=10), H-domain-dist (epi<edu), H-catalog-flip-prediction (flipPred=54.1%), H-domain-flip-geometry-controlled (econHighImbFlips=10), H-econ-encoding-selection (mismatch=50%), H-domain-profiles-stable (econFlipRate=25.6%). H-catalog-exact-match still PASS (39.1%). Five theorems + all critical invariants unchanged. Corpus patterns are scale- and selection-contingent; laws are not.

Loop replay: [loop-78]

conc-expansion-wave-synthesis · Methodology

Expansion waves 1–2 (loops 68–72): adversarial stress tests survived; theorem candidacy patterns held or sharpened

Pending loop-72 LoopReview. Witnesses collider-no-manifest, economics encoding selection, sweep-fragile class, domain profile stability at N>110.

Loop replay: loop-76

Scope / open question (2)

conc-08-type-c-real · Scope / open question

Does the published literature contain real type-C studies (compression without sign flip)?

Witnessed in loop-24: hannan-1994 (CABG surgery), titanic-1912, melanoma-altman-1991, and coffee-tverdal-2020 all hydrate as DistortionType=C. AllocationDistortion>0.01, IsSignFlip=FALSE — the correct pooled direction but compressed magnitude. The type-C cell is not a synthetic pathology.

Loop replay: loop-24

conc-09-causal-vs-geometric · Scope / open question

AllocationDistortion is a geometric measure; causal interpretation requires external domain knowledge, gated by AdjustmentAppropriate

Berkeley has the highest AllocationDistortion (0.1934) but CausalRole=contested (mediator vs confounder), proving that geometric severity and causal confounding are not the same thing. The instrument is deliberately geometric: it classifies distortion type and flags the corrected winner from allocation arithmetic alone. Whether it is safe to act on that correction as a causal claim is answered by AdjustmentAppropriate — which gates on ConditioningRisk and CausalClaimStatus. The two layers are complementary, not …

Loop replay: loop-30
Protecting invariants: 1

Observed in this corpus (2)

conc-01-paradox-is-derived · Observed in this corpus

Simpson's Paradox is a derived fact from the DAG, not a modeled entity

IsReversal (and later IsSignFlip) falls out as a calculated field on TreatmentRankings, derived from aggregated CaseCell counts. No 'ReversalDetection' entity was ever needed.

Loop replay: loop-04
Protecting invariants: 2

conc-02-reversal-from-allocation · Observed in this corpus

The reversal in kidney-1986 is entirely an allocation artifact — same rates with balanced allocation produce no reversal

kidney-balanced uses identical stratum success rates to kidney-1986 but with equal allocation (175 cases per cell). IsReversal=FALSE, AllocationDistortion=0. The paradox is algebraically isolated as a property of allocation, not of treatment efficacy.

Loop replay: loop-14
Protecting invariants: 3

Invariant health

Critical invariant failures: 0
Definitional (cannot fail by construction — see definitions above): 22
Judgment-crossing (touch hand/LLM-encoded CausalRole): 4
Vacuous (0 pass / 0 fail — empty universe, proves nothing yet): 0

inv-import-session-ready: Loop-67+: expansion backlog loaded; bulk encode session ready when >=20 candidates and >=15 encode-ready. (fail=1)

inv-pooled-gap-crosses-zero: Legacy sweep check from ratio-based Type A (pre-loop-70). Type A is now unanimity-based; cross-zero is informative but not definitional. (fail=1)

inv-pooled-gap-wanders: Pooled gap should move under allocation reweighting for reversal studies. Warning-only: small-magnitude reversals may have narrow sweep range. (fail=1)

inv-type-a-ratio-negative: Legacy ratio geometry check (pre-loop-70). Type A is now IsStratumUnanimous; ratio band retained as warning-only witness. (fail=4)

inv-type-b-ratio-lt-neg1: Legacy ratio geometry check (pre-loop-70). Type B is now heterogeneous stratum directions; ratio threshold retained as warning-only witness. (fail=2)

Judgment-crossing invariants (the closest thing here to a substantive check):

inv-collider-no-manifest: Real-data collider and selection studies never produce manifest sign-flips at observed allocation — conditional theorem requiring correct causal-role annotation (loop-75). Scoped to real-data studies: loop-89's collider-purity-boundary-flip is a deliberately-constructed synthetic adversarial stress test (1.2M-trial search for the highest-purity manifest flip a collider geometry can produce, purity=0.49994) built specifically to probe this theorem's boundary from the outside — it is evidence the boundary is real and tight, not a counterexample to the real-data claim (loop-92). (pass=11, fail=0)

inv-confounder-signflip-explained: Every confounder manifest sign-flip must be marked explained. Other half of the explained!”confounder biconditional (loop-74). (pass=85, fail=0)

inv-explained-sign-flip-confounder: Every IsParadoxExplained=TRUE row must be a confounder manifest sign-flip. Half of the explained!”confounder biconditional (loop-74). (pass=85, fail=0)

inv-unexplained-identity-sink: Every real-data unexplained sign-flip must be linked to contested-org-choice or collider-proxy identity — no unexplained flip belongs to an unambiguous confounder archetype. Identity-conditioned extension of conc-29 (loop-74 biconditional). Scoped to real-data studies (loop-92): loop-89's collider-purity-boundary-flip is a deliberately-constructed synthetic study with no StratumVariableIdentityMaps row at all (synthetic studies are never identity-mapped), so it is not a counterexample to this real-corpus claim. (pass=3, fail=0)

Most misleading studies (by paradox strength)

COVID-19 CFR: Italy vs China by Age (von Kügelgen 2021) — type C+, strength 0.2743, distortion 0.529, purity 3.6% (CAUTION)
Triathlon On-Time Finish: Sex by Age Cohort — type A, strength 0.1455, distortion 0.245, purity 28.9% (DANGER)
Smoking and Mortality by Age (Cochran / Surgeon General 1964) — type A, strength 0.1091, distortion 0.167, purity 25.7% (DANGER)
UC Berkeley Graduate Admissions (Bickel et al. 1975) — contested confounder — type B, strength 0.0969, distortion 0.193, purity 24.9% (DANGER)
UC Berkeley Admissions: Six Largest Departments (Bickel et al. 1975) — type B, strength 0.0944, distortion 0.184, purity 18.8% (DANGER)
Exercise and Cholesterol Control by Age Group (Framingham-style) — type D, strength 0.0800, distortion 0.008, purity 95.2% (SAFE)
Baseball Batting Averages: Lefebvre vs Fairly World Series (Day 1994 / baseball paradox canon) — type A, strength 0.0606, distortion 0.082, purity 20.9% (DANGER)
Whickham Smoking Cohort — 20-year Mortality by Smoking Status (Appleton et al. 1996) — type B, strength 0.0538, distortion 0.119, purity 26.9% (DANGER)

Rulebook & repo (full derivation)

Formulas, Loops build history, InvariantChecks, and Conclusions rows live in the rulebook JSON. README.md carries the full framing.

Project folder: https://github.com/EffortlessAPI/effortless-rulebooks/tree/main/rulebook-examples/simpsons-paradox
README (full repo guide): README.md

Rulebook SSoT (formulas, Loops, InvariantChecks, Conclusions): effortless-rulebook/simpsons-paradox-rulebook.json
Rulespeak (plain-English business rules): rulespeak/rulespeak.md
Postgres substrate (views, init-db.sh): effortless-postgres/README.md
Standalone HTML explorer: simpsons-paradox-explorer.html
Live API (with ./start.sh): GET localhost:3001/api/invariant-checks · /api/treatment-rankings · /api/model-summary